

Where R_k is the vector of portfolio shares in period k , this reduces to:

$$\mathfrak{R}(n) = \sqrt{\sum_{k=1}^n R'_k C R_k} \quad (7.33)$$

Trading risk for a trade schedule for a single stock execution is calculated as follows:

$$\mathfrak{R}^2(r_k) = \sum_{j=1}^n r_j^2 \cdot \sigma^2 \cdot P_0^2 \quad (7.34)$$

where σ^2 is the corresponding one-period variance expressed in $(\$/\text{Share})^2$, P_0 is the current price, and $\mathfrak{R}^2(r_k)$ is the total dollar variance for the strategy. Notice that we are simply summing the variance in each period.

For a continuous trade rate strategy where we execute the same number of shares in each period, the number of unexecuted shares at the beginning of each trade period is calculated as follows:

$$r_j = X - \frac{X}{n} \cdot (j-1) = X \left(1 - \frac{(j-1)}{n} \right) \quad (7.35)$$

where X is the total number of shares in the order.

Then the number of unexecuted shares at the beginning of each period squared is:

$$r_j^2 = X^2 \left(1 - \frac{(j-1)}{n} \right)^2 \quad (7.36)$$

Now let,

σ^2 = the annualized variance

t^* = total time to trade in terms of a year (same units as volatility),

e.g., $t = 1$ is one year, $t = 1/250 = 1$ day, etc.

n = number of periods in the trading interval

Then we have,

$t^* \cdot \sigma^2$ = variance scaled for the time period

$t^* \cdot \frac{\sigma^2}{n}$ = variance scaled for a trading interval

For example, if the trading time is one day and the day is segmented into ten periods then we have:

$$\sigma^2(\text{trading period}) = \frac{1}{250} \cdot \frac{\sigma^2}{10} \quad (7.37)$$